

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Fractions

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

FRACTIONS · CH1 · L5

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The Map

This lesson collapses Fractions into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Use a common denominator

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Multiply and divide fractions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Simplify mixed numbers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Convert the final form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Simplify fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Convert to improper fractions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 14 website questions, 14 checked solutions, 3 local PDFs read.

01 Use a common denominator

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4\frac{2}{3} = \frac{14}{3}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q3

Convert to improper fractions

$$4\frac{2}{3} = \frac{14}{3}$$

Use a common denominator

$$\frac{14}{3} + \frac{19}{5} = \frac{70}{15} + \frac{57}{15}$$

Finish: $8\frac{7}{15}$

FORMULA BANK

Common denominator for addition/subtraction. Multiply straight across. Divide by the reciprocal.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Multiply and divide fractions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3\frac{1}{5} = \frac{16}{5}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q2

Simplify fraction

$$3\frac{1}{5} = \frac{16}{5}$$

Simplify fraction

$$\frac{16}{5} \times \frac{21}{8} = \frac{2 \times 21}{5}$$

Finish: $8\frac{2}{5}$

FORMULA BANK

Common denominator for addition/subtraction. Multiply straight across. Divide by the reciprocal.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Simplify mixed numbers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3\frac{2}{3} = \frac{11}{3}, \quad 2\frac{4}{7} = \frac{18}{7}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2025 4WM1H Q1

Simplify fraction

$$3\frac{2}{3} = \frac{11}{3}, \quad 2\frac{4}{7} = \frac{18}{7}$$

Simplify fraction

$$\frac{11}{3} \times \frac{18}{7} = \frac{198}{21} = \frac{66}{7}$$

Finish: $9\frac{3}{7}$

FORMULA BANK

Common denominator for addition/subtraction. Multiply straight across. Divide by the reciprocal.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

04 Convert the final form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$6\frac{3}{4} = \frac{27}{4}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2H Q2

Convert to improper fractions

$$6\frac{3}{4} = \frac{27}{4}$$

Divide by multiplying by the

$$\frac{27}{4} \div \frac{18}{7} = \frac{27}{4} \times \frac{7}{18}$$

Finish: $2\frac{5}{8}$

FORMULA BANK

Common denominator for addition/subtraction. Multiply straight across. Divide by the reciprocal.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Simplify fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3\frac{1}{5} = \frac{16}{5}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q2

Simplify fraction

$$3\frac{1}{5} = \frac{16}{5}$$

Simplify fraction

$$\frac{16}{5} \times \frac{21}{8} = \frac{2 \times 21}{5}$$

Finish: $8\frac{2}{5}$

FORMULA BANK

Common denominator for addition/subtraction. Multiply straight across. Divide by the reciprocal.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 Convert to improper fractions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4\frac{2}{3} = \frac{14}{3}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q3

Convert to improper fractions

$$4\frac{2}{3} = \frac{14}{3}$$

Use a common denominator

$$\frac{14}{3} + \frac{19}{5} = \frac{70}{15} + \frac{57}{15}$$

Finish: $8\frac{7}{15}$

FORMULA BANK

Common denominator for addition/subtraction. Multiply straight across. Divide by the reciprocal.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Use a common denominator
"work out", "show", "hence", a familiar exam phrase	Multiply and divide fractions
"work out", "show", "hence", a familiar exam phrase	Simplify mixed numbers
"work out", "show", "hence", a familiar exam phrase	Convert the final form
"work out", "show", "hence", a familiar exam phrase	Simplify fraction
"work out", "show", "hence", a familiar exam phrase	Convert to improper fractions

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.